

Update Day 3 – March 31st 2021

Location: Online (Pitch2Peer, BB, Academic Professional)

PPT-presentations (Videos)	
Karimi	Anahid
Bresser	Soraya
Buil	Simon
Campisano	Carlotta
Daniels	Andreas
Deetman	Maria
Fanlo Garcia	Sònia
Kieran Mc Donagh	John
Klapdor	Jule
Klein Hazebroek	Bente
Klomp	Maaïke
Kuik	Kimberley
Woods	Tom

Poster-presentations (poster & written pitch)	
Anastasopoulou	Theofania
Bergmans	Anne
Brasser	Mariëlle
Driessen	Esti
Franssen	Joep
Guldenaar	Nadia
Nuesink	Satyana
Pelsma	Iris
Prummer	Franziska
Pruthi Kola	Sara
Schouten	Lisanne
Sheedy	Sarah
Siemensma	Friso
Simon	Clara
van den Esschert	Floris

PPT-Presentations

Name Student: Anahid Karimi

Internship or Thesis: Internship

Title project: How gaining situational awareness affects take-over readiness in semi-automated vehicles: a predictive model.

Objective: I present a mathematical model of take-over readiness in semi-automated driving. The model simulates how a linear way of gaining situational awareness impacts take-over readiness. **Background:** As the driving industry gradually shifts from manual to (semi-) automated vehicles, more research on safety risks and human-machine interaction is needed. Previous studies proposed frameworks for the stages that occur when control over a vehicle is transferred from vehicle to driver, and which factor play a role. However, less research has focused on how drivers move through the stages of interruption and what this means for take-over readiness. **Method:** A model was implemented in Excel. As a proof-of-concept, I use the model to compare three example scenarios of take-over readiness. **Application:** This model offers a first step towards understanding the take-over process in semi-automated driving and can contribute to the design of safe and effective semi-automated vehicles.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: dr. Chris Janssen

Name Student: Soraya Bresser

Internship or Thesis: Thesis

Title project: The effect of oxytocin on eye contact and pupil diameter in patient-physician interaction.

The aim of current study was to understand whether enhanced levels of oxytocin lead to increased eye contact and increased pupil diameter in patient-physician interaction. Furthermore the relationship between eye contact and pupil diameter was studied, to see whether there exists a positive relationship as well. Oxytocin is a hormone known for its important role in human attachment and social behaviour. Previous studies on the influence of oxytocin on eye contact and pupil diameter, as well as studies on the relationship between eye contact and pupil diameter have not been conducted using the patient-physician context. This can be of specific interest as the social salience hypothesis assumes oxytocin's effect on human social behaviour to be context dependent. Eighty-one healthy young-adult males participated in this randomized double-blind placebo-controlled crossover trial. After receiving oxytocin/placebo, participants imagined themselves being the patient, while watching the physician in a video. Eye contact and pupil diameter were measured using the SMI RED 500 eye-tracker.

Name company or University: Amsterdam UMC

(If applicable) External supervisor: Chiara Jongerius

UU Supervisor: Yentl de Kloe

Name Student: Simon Buil

Internship or Thesis: Internship

Title project: The flaws in AGL-studies

My internship was about the way in which animals can learn syntax like rules. They developed artificial grammars to test these abilities, but there is one flaw present in many of the AGL-studies. The flaw is that there are more acoustic similarities in correct strings than there are similarities in the incorrect strings. During my internship I tested 15 AGL-studies on their amount of acoustic similarities divided over 4 categories (contains, shared start, shared substring and cross-correlation). I found that only 2 out of the 15 passed the test of acoustic similarities and we can thus conclude that future AGL-studies should focus on overcoming the flaw of acoustic similarities and that the moment we can not conclude much over the grammar ability of animals.

Name company or University: Utrecht University

UU Supervisor: Gabriel Beckers

Name Student: Carlotta Campisano

Internship or Thesis: Internship

Title project: Excel-lent Member Tracking

AIESEC International is a volunteer work organization that promotes international contact and leadership in the youth. AIESEC is active in more than 108 countries all over the world. Two types of people interact with AIESEC: the members, all aged around 18-30, who manage every aspect of the organization, and the customers, who are interested in paying for an experience of volunteer work or internship abroad. My work within the organization was concerned with the members, and not the customers. The aim of my internship was to answer the following two questions.

1) What bring some members to leave the committee? Are there any recurring patterns?

2) How can we re-structure the Human Resources tracker to facilitate the process of data collection and analysis?

The first two questions were answered through the analysis of member data available to me as of November 2020, while the last question was addressed by creating a new tracker suited to easily collect data and automatically compute descriptive statistics.

Name company or University: AIESEC in Catania

(If applicable) External supervisor: Antonio Messina

UU Supervisor: Chris Paffen

Name Student: Andreas Daniels

Internship or Thesis: thesis

Title project: The discreet use of ASMR in video commercials to improve brand experience

My presentation is about my Master Thesis. In my presentation I will explain my thought process, like how I came up with my thesis idea and how I made my experiment in Qualtrics. My thesis is about ASMR, so I will explain what ASMR exactly is and I will use some sounds to give examples how ASMR sounds like. I implemented a stimuli that I used in my experiment. I show this video in the beginning of my presentation and at the end. I hope that people will notice the ASMR in this video at the end of my presentation. I suppose that people will not hear it at the beginning, because no one that did my experiment heard ASMR.

Name company or University: Utrecht University

UU Supervisor: Anouk Keizer

Name Student: Maria Deetman

Internship or Thesis: Thesis

Title project: Attitudes and expectations towards social robots in children's music education

In music education, practicing is an important aspect. It can, however, be hard to stay motivated, especially for children. The presence of others can improve motivation and increase performance and social robots can elicit the same effect. For robots to help young students in this way, parents and teachers need to accept a robot in this role. In this study, parents and teachers are asked about their attitude, expectations and intentions toward using social robots. Results showed that intention depends on their attitude towards robots in music education specifically, more than their attitude towards robots in general. Participants expect robots to give positive feedback and schedule and structure practice sessions. Moreover, parents expect the robot to notice mistakes made by the student. The biggest concerns are the robot's inability to convey the emotional aspects of music or to demonstrate instrument playing. These results can help designers improve robot designs.

Name company or University: Eindhoven University of Technology (TU/e)

External supervisor: Heqiu Song

UU Supervisor: Roy Hessels

Name Student: Sònia Fanlo Garcia

Internship or Thesis: Thesis

Title: The effects of light probe's physical characteristics on image light perception

Light probes are spherical objects that help infer light in space by taking cues from the probes' lighting characteristics. We tested whether differences in material, shape and surface structure of light probes influenced the ability of observers to gather information about lighting in natural settings. We compared three light probes: A white matte sphere, a black shiny sphere and a golf ball. Participants were shown images of a probe photographed in a scene. There were 13 original combinations of scene and probe. Some images had the original probe photographed in the scene, but in other images the probe was swapped for that of a different scene. Participants had to determine if the probe fitted the scene. The main implications of this project are to consolidate light probes as a tool for designers to present their projects and provide guidelines on which light probe to use for a specific scene.

Name company or University: Universiteit Utrecht

UU Supervisor: Susan Te pas

Name Student: John Kieran-Mc Donagh

Internship or Thesis: Thesis

Title project: The Cognitive Effects of Webcam Surveillance.

A side effect of the recent pandemic is that we now often view our family, friends and colleagues through the lens of a webcam. However, relatively little is known about the cognitive, behavioural and psychological effects that they may induce (Claypoole & Szalma, 2019). In this study, we aim to

look at webcam surveillance monitoring, part of a pool of literature known as electronic performance monitoring, (Ravid, 2020). This is achieved by providing little or no rationale for the purpose of the webcam. We hypothesize that webcams will elicit similar effects to social presence, and restrict participants' attention capacity, resulting in a decrease in cognitive performance. A total of 82 participants completed an online recognition-memory (RM) task and Stroop task. Half of the students were told they were being recorded, while the other half were not. Linear mixed effects models revealed a significant congruence effect, and congruence x accuracy effect ($p < .001$), as a result of webcam presence. Participants were significantly slower at responding to incorrect congruent trials in the webcam condition than in the non-webcam condition. However, not enough evidence was accumulated to accept our main hypothesis. Theoretical implications and practical applications are further discussed.

Name company or University: Utrecht University

UU Supervisor: Dr. Leendert van Maanen

Name Student: Jule Klapdor

Internship or Thesis: Thesis

Title project: Creativity enhancing Virtual Reality work environments

In recent years, companies have increasingly focused on innovative work environments to foster the creative potential or performance of their employees (Google/Apple). Due to the current Covid 19 pandemic, more and more people are working from home and companies are faced with the challenge of creating a working environment that promotes creativity in the private sphere as well. In this context, the use of Virtual Reality (VR) can be used to promote creative processes. Within my thesis, a virtual, creativity-enhancing working environment is generated and tested in the lab. We are comparing two environments, manipulated in their degree of abstractness. Concrete is defined as a closed environment and abstract is defined as a vast environment. We hypothesize that the abstract environment enhances creativity. We also test whether awe moderates the relationship. To increase ecological validity, we are measuring individual and dyadic (two persons interact) creativity performance and subjective creativity.

Name company or University: University of Würzburg/ Fraunhofer Institute/ Loughborough business school

External supervisor: Carolin Wienrich & Brid Karacan

UU Supervisor: Ruud Hortensius

Name Student: Bente Klein Hazebroek

Internship or Thesis: Thesis

Title project: Communicative Coffee: Exploring the Effect of Coffee Flavour Descriptions on Consumer Behaviour

In my thesis, I will explore how coffee flavours can be communicated to regular coffee consumers. Coffee knows a broad range of flavours, so it is useful if coffee producers include a short flavour description on bags of coffee. A product description can shape consumers' expectations, and, with that, their behaviour. One of the things that consumers constantly do when choosing a product, is considering potential risks, and one of such risks is a product's flavour. Consumers want to know what a product tastes like if they have not tried it before. Even when there is a broad range of coffee

flavours, and there is little agreement between speakers, descriptions on coffee packaging should be understandable for regular consumers. Experiment I explores to what extent the type of words to describe coffee flavour (concrete/abstract/evaluative words) influences consumer behaviour, and experiment II explores to what extent the structure of coffee flavour descriptions (stative/sensory/figurative narrative) influences consumer behaviour. The experiments are online questionnaires, in which participants see online advertisements of a bag of coffee with manipulated flavour descriptions, and they are asked to assess consumer behaviour on the three levels gustatory imagery, desire to taste, and willingness to pay. An ANOVA will be conducted to see if there are significant differences between the conditions.

Name company or University: Utrecht University

UU Supervisor: Ilja Croijmans

Name Student: Maaïke Klomp

Internship or Thesis: Thesis

Title project: The relation between a Parkinson patients' a priori information needs, information provision by neurologists, and patients' retrospective experience of information received during diagnostic consultations

Abstract: My thesis is part of the DiaMove research project at the Amsterdam Medical Center and is about the relation between a Parkinson patients' a priori information needs, information provision by neurologists, and patients' retrospective experience of information received during diagnostic consultations. Recent research indicates that patients retrospectively report to feel insufficiently educated about their disease during diagnostic consultations. The exact cause of this is unknown. My thesis focuses on how information provided during diagnostic consultations for Parkinson's disease (PD) matches the information reported afterwards by patients and tries to find what causes the patients reporting to be not fully informed. The knowledge gained from this research will contribute to optimizing diagnostic and post-diagnostic consultations about PD, which may ultimately contribute to improved patient well-being, satisfaction and trust.

Name company or University: Amsterdam UMC

External supervisor: Liesbet Kurpershoek

UU Supervisor: Elze Wolfs

Name Student: Kimberley Kuik

Internship or Thesis: Thesis

Title project: *The Relationship Between Arousal and Cognitive Control and its Modulation by Trait Arousability*

Over the past years, there has been increasing attention to the replication crisis in psychological research. In 2015, the Open Science Collaboration tried to replicate 100 psychological studies, and found that only one third to half of the original findings could be replicated. Since replicability is a big aspect of Open Science, it is important that the reproduction rate is increased. That is why I will try to replicate the findings from a study by Demanet, Liefooghe and Verbruggen (2011) for my thesis. I will create an online follow-up study, investigating the relationship between arousal and cognitive control. As an add-on component, I will look at how this relationship is modulated by trait arousability. It is expected that arousal affects cognitive control, as is found in the original

experiment.

Name company or University: Utrecht University

(If applicable) External supervisor: N.A.

UU Supervisor: Baptist Liefoghe

Name Student: Tom Woods 7361068

Internship or Thesis: Thesis

Title project: Saving Cash: A quantitative analysis of the relationship between loss aversion and payment methods.

There has been little quantitative analysis of the relationship between loss aversion and payment mode. The purpose of this study was to probe any differences in use of physical and digital money, measuring these differences on a scale of loss aversion. This analysis was carried out through a series of low-stake gambles with an Irish population. Participants were incentivised by use of a competitive game. A 2 x 3 between subject ANOVA was conducted to test whether people are more loss averse with physical or digital payment mechanisms. It was found that individuals, specifically males, took significantly higher risks when using digitally money, over physical money ($p < .05$). The results of this research suggest that the move towards becoming a cashless society impacts human behaviour. It can be seen by higher levels of risk being accepted in the digital condition, particularly by the male population.

Name company or University: Utrecht University

UU Supervisor: Martijn Mulder

Poster-presentations

Name Student: Theofania Anastasopoulou

Internship or Thesis: Internship

Title project: *Eye tracking: a study of head movements by video gamers*

Eye tracking manufacturers market their products as able to handle head movements, providing large accuracy and precision in optimal conditions. That means that the user's head needs to be fixed or restricted in the original position inside a headbox (range of distances to eye tracker); as it moves from it the data quality drops dramatically. Tobii Pro, a leading eye tracker company, has supported that they manufacture their gaming eye trackers based on the notion that video gamers do not move their head during gameplay and that their headbox can support a variety of head movements. Based on this, I wanted to explore whether they actually make head movements (first hypothesis) and if they are affected by the type of the game (second hypothesis). To test this, a behavioral coding scheme with head movements and their explanations was developed. Streamers for three different games were observed: Call of Duty (COD), League of Legends (LOL) and Hearthstone (HS), by three observants (coders). The results showed movements in all games, confirming the first hypothesis, as well as differences in numbers between them, but they were not significant. The type of the game did not influence the number of head movements. Based on the headbox dimensions and the literature, coded movements such as "Big", "Fixed" or "Slow Drifting" can have a big impact on the eye gaze data quality.

Name company or University: Utrecht University

UU Supervisor: Ignace Hooge

Name Student: Anne Bergmans

Internship or Thesis: Internship

Title project: On Your Feet! The combined effects of outdoor walking and errorless learning interventions on physical fitness, cognitive functioning, quality of life and learning in Korsakoff patients.

During my internship at Korsakoff Clinic Slingsdael in Rotterdam I was responsible for one of the research projects focused on optimizing overall quality of life and independence of inpatients with Korsakoff syndrome. The project was focused on assessing the possible benefits of combining two types of existing interventions (a physical and a learning intervention) and contributing to the currently very scarce intervention based literature on Korsakoff syndrome. Findings are very promising and show Korsakoff patients are still able to learn a new route, despite very prominent memory problems, and that learning may be enhanced by certain learning techniques. Furthermore, very short and low intensity physical interventions already prove to be beneficial in enhancing physical wellbeing of patients with Korsakoff syndrome. A recording of the presentation for the staff of the Korsakoff clinic about my research and findings can be found under the QR code on the right (in Dutch).

Name company: Slingsdael Korsakov Kliniek (Lelie Zorggroep)

External supervisor: Erik Oudman

UU supervisor: Albert Postma

Name Student: Mariëlle Brasser

Internship or Thesis: Thesis

Title project: Presence in Microsoft Teams

During the corona pandemic a lot of our face-to-face interactions have been replaced by online videoconferencing. Microsoft has now developed a new mode for their videoconferencing program Microsoft Teams, namely 'Together mode'. Amongst others, Microsoft's claim for this program is that it makes it feel like everyone is in the same room. In other words, Together mode has a higher sense of presence -the subjective feeling of being in a virtual space- than normal grid view. In this research I investigate if that is really the case, since Microsoft hasn't published any proof on this. I designed an experiment in which one group discussed some topics in grid view, and another group in Together mode. Afterwards both groups filled in a questionnaire and results from this questionnaire were compared with an independent-samples t-test. Presence scores in Together mode were not significantly higher than in normal grid view.

Name company or University: Utrecht University

UU supervisor: Roy Hessels

Name Student: Esti Driessen

Internship or Thesis: Thesis

Title project: The MADS game: Is it effective?

IJsfontein has developed a game to combat obesity in young people: the MADS game. In this game, young people will learn behaviors that ensure a healthier lifestyle. The effectiveness of the game is

examined by means of a pilot study. About 50 overweight young people will try out the game. To determine the effectiveness of the MADS game, the comprehensibility of the app as well as the user experience are measured, by means of surveys, interviews and the user data. In addition, the difference in lifestyle and self-determination at the beginning and at the end of the pilot will be determined by using a paired t-test.

Name company or University: IJsfontein

(If applicable) External supervisor: Annelies Wisse

UU Supervisor: Caroline Junge

Name Student: Joep (JJI) Franssen

Internship or Thesis: Internship

Title project: Developing a location independent testing procedure for the Neurolympics

Abstract: During my internship at BrainsFirst (26 October - 29 January), I have worked on the development of a new test protocol for the Neurolympics. The Neurolympics is a series of online games that measure someone's cognitive skills. In the past, a BrainsFirst instructor visited soccer clubs and supervised the players while they were playing the games in the same room. Due to Covid-19 and requests from clubs abroad, BrainsFirst was increasingly interested in developing a location-independent testing procedure. Besides, with the old procedure, some players did not listen carefully to instructions or started annoying other players in the room resulting in a low data quality. The newly developed testing procedure relies entirely on a digital infrastructure. The players can do the Neurolympics from home as well as the BrainsFirst supervisor. The players get instructions through an online meeting and play the games using a game tips website at their own pace. If they need help, they can reach out through a live chat for personal support. Moreover, the data quality was better during the trials than with the old procedure. Using recording software, active webcams and good instructions, we control their environments to get the best data quality. Given the promising results of the new test protocol, it is currently being used for top-tier clubs like Feyenoord, AZ and the Royal Dutch Football Association (KNVB).

Name company or University: BrainsFirst

(If applicable) External supervisor: Mike Keijser

UU Supervisor: Dr. Caroline Junge

Name Student: Nadia Guldenaar

Internship or Thesis: Internship

Title project: From Nudge to Support

Young adults are often not actively engaged in giving to charity, which means that a big part of the society is often 'left out'. Nevertheless, their help is essential, especially for organisations such as KNGF Geleidehonden: Their income comes mainly from individuals, rather than receiving subsidised income. Therefore, this study focusses on psychological determinants on why young adults do or do not give to charity. A literature research has been conducted and a practical advice is drawn upon the performed literature research. According to the findings, one could assume that young adults find it hard to act upon their good intentions to donate money to charity organisations.

Furthermore, young adults seem to differ from older adults and elderly in the way they want to be approached: Their costs should be reduced as much as possible and they are especially sensitive to

social influences. In order to facilitate acting upon their good intentions, this study advises KNGF Geleidehonden to reach young adults by including nudges in their campaigns; nudges facilitate translating good intentions into desired behaviour. As young adults seem to be especially sensitive to social influences, social norm nudges and different framing techniques are considered. This advice has been tested via social media; the results suggest that this advice does work in the field.

Name company: KNGF Geleidehonden

External supervisor: René Keijer

UU Supervisor: Michiel Vellema

Name Student: Satyana Nuesink

Internship or Thesis: Internship

Title project: Working at home after the COVID-19 pandemic

Between 45 and 60 percent of employees intend to continue working at home (part time) after COVID-19, which has a positive effect on traffic flow and accidents. For employers this is desirable because, among other things, they save money on office space and employees are more flexible and absent less often. However, because the action of 'going to the workplace' is for many people a strong habit, it cannot be assumed that these intentions automatically predict behaviour, without extra stimulation. Therefore, the current research investigated: *"What can employers do right now if they want to stimulate their employees to continue to work at home (part time) after the COVID-19 pandemic?"* The research question was answered with a literature study. Theories on motivation, intention, habit, and behaviour change were discussed in the context of working at home. Important elements were combined into five categories of strategies: (1) positive reinforcement, (2) autonomy, (3) thinking through and planning behaviour, (4) leadership, and (5) implementing lost values. Within these categories, twelve corresponding concrete advices for employers were formulated. It is hypothesized that if the concrete advices relating to these categories are actualized within a company, than more of its employees will remain to work at home (part time) after the COVID-19 pandemic is over.

Name company or University: Goudappel

(If applicable) External supervisor: Matthijs Dicke-Ogenia

UU Supervisor: Dennis Schutter

Name Student: Iris Pelsma

Internship or Thesis: Internship

Title project: Increasing conversion rate: Target the right audience with right content.

Increasing the conversion rate; more purchases made on a website. It is the most important service a marketing agency as Elevate Digital can provide. To do so, it is important to gain information about the right audience. Since these insights can be used in future research questions concerning personalisation.

Personalisation is generating specific content for the target audience.

During my internship I was responsible for the personalisation track of webshop Pip Studio and the aim was to provide new insights for them that led to more conversion and strengthen the literature.

For Pip Studio I investigated the target audience, created content for this audience and tested these experiments to conclude if it led to an increase of the conversion rate.

The results show that tests based on the persuasion principles of Cialdini at which the target audience was susceptible for led to an increased conversion rate of 29,77% . Besides, tests that adressed the buyer type led to an increased conversion rate of 24,50%. At last, the test that adressed the goals of the target audience led to an increased conversion rate of 22,40%. Each webshop has a different audience. Therefore it is important to investigate this target audience in advanced and to set up tests that fit that audience. For the webshop Pip Studio this research has found that persuasion principles, type of buyer, and their goals led to an increase of the conversion rate. These insights can be used to set up tests for other customers of Elevate Digital.

Name company or University: Elevate Digital

(If applicable) External supervisor: Bart Koomen

UU Supervisor: Nico Romeijn

Name Student: Franziska Prummer

Internship or Thesis: Internship

Title project: Face Masks and Emotion Perception

The use of face masks throughout the COVID-19 pandemic has become a common sight to reduce the virus spread. However, this approach covers the mouth and nose region of the face, impairing emotion perception. A pilot study was set up to investigate a possible relationship between face masks and the ability of correctly identifying emotions. We asked 13 participants between the ages 23-53 to conduct a two-alternative forced choice task. The experiment stimuli consisted of masked and unmasked non-binary gender faces transitioning from a happy to angry emotion at different intensity increments. Participants were asked to label faces as either “friendly” or “unfriendly”. We found an influence of masks on the performance of correctly identifying happy emotions. It was more common to falsely label happy emotions as “unfriendly” than contrariwise. These results suggest that face masks cause people to detect friendly emotions more negatively.

Name company or University: Utrecht University

UU Supervisor: Martijn Mulder

Name Student: Sara Pruthi Kola

Internship or Thesis: Thesis

Title project: A dual eye tracking study: Uncovering gaze patterns during live conversations with different emotional content

The importance of the gaze has mesmerized writers, psychologists, philosophers and body language experts for ages. Despite the importance of social context, most studies that investigate gaze behavior have employed pictures of faces or social scenes, thereby ignoring the possible dynamics in gaze behavior when social partners are physically present. Hence, the aim of this study is to allow social interaction between participants while recording their gaze. What we want to find out is which facial features do people gaze at when in interaction with each other in conversations with different emotional content and how does the gaze behavior of one participant relate to the gaze behavior of the other participant? To answer this, we designed an experiment where the participants were asked to converse about pictures with different emotional content, while their gazepath was being recorded by two Tobii eyetrackers. Hereupon, the data is being analyzed with

Rstudio.

Name company or University: Utrecht University

UU Supervisor: Niilo Valtakari

Name Student: Lisanne Schouten

Internship or Thesis: Internship

Title project: Feeling well (informed)? *A literature study and survey study on education about psychological side-effects of the contraceptive pill.*

The contraceptive pill is an important medicine used by over 100 million women around the world. Even though a lot of research has been conducted about its physical side effects and risks, a lot is still unclear about the psychological effects of the pill. Uncertainty about this could lead to unsubstantiated fear towards and aversion to the pill. In this research an answer was sought to the question: 'How can education about the psychological effects of the pill be improved?'. The aim was to develop some general guidelines to assist general practitioners in providing complete and understandable information about psychological effects of the pill. In a first study, existing literature about psychological effects was shortly reviewed and the most important information was determined. In a second study, results of a survey about pill education among over 300 former and current Dutch pill-users were discussed. Most substantiated psychological pill effects were the potential impediment of unlearning fear and anxiety disorder recovery, libido decrease and contribution to depressive symptoms. Information women wanted to be educated about was that about hormonal functioning of the pill, how hormones and mood and depression are related, about positive and negative mood effects, including the risk for depression, about disadvantages of the pill, different pill types and alternative contraception methods.

Name company or University: Utrecht University, Psychology Department

UU Supervisor: Dr. Estrella Montoya

Name Student: Sarah Sheedy

Internship or Thesis: Internship

Title Project: A Nieuw Initiatief: Discovering the Importance of User Experience

The objective of this internship is to assist Nieuw Initiatief by utilize psychological knowledge to implement changes and create new systems that optimize the user's experience. Nieuw Initiatief, a user experience company, focuses on creating digital solutions for companies by administering usability tests, designing websites, and creating e-learning or online communities. For this project various tasks were executed in order to apply relevant knowledge to the issues that Nieuw Initiatief was facing. One issue plaguing the company was trying to understand the user journey of the new online community and e-learning platform in order to create a design that was best suited for the target audience. In addition, new perspectives were necessary in designing a new website as well as composing a template for social media content. Finally, the primary project of this internship was to design a usability test to offer to the customers of Nieuw Initiatief. Each task was done in a sequence that allowed insight and knowledge to be gained and utilized to assist with the making of the usability test. Below, each of the completed tasks will be described while also addressing the initial issues and their importance for Nieuw Initiatief.

Name Company or University: Nieuw Initiatief

External Supervisor: Edwin de Kuiper

UU Supervisor: Stella Donker

Name Student: Clara Simon

Internship or Thesis: Internship

Title project: What's in a glimpse? An investigation of the OutSystems beginner tutorial with webcam eye-tracking

In my internship, I presented the state of the art webcam eye-tracking technologies and researched their usefulness in order to get insights about the usability of the OutSystems product. Since online usability research offers great flexibility and access to a wide range of participants, webcam eye-tracking becomes relevant as a supporting data source to usability studies. As a first approach, the RealEye eye-tracking algorithm was applied to the OutSystems beginner tutorial. The usability of the eye-tracking platform itself was investigated. Moreover, I evaluated if the quality of measurements is suitable to get insights about the tutorials' usability and how participants were able to adapt to an eye-tracking friendly environment. We concluded that this research method is suitable for future application and integration in the OutSystems UX research toolkit, as long as some constraints are considered.

Name company or University: Utrecht University/OutSystems

(If applicable) External supervisor: Carina Fonseca

UU Supervisor: Krista Overvliet

Student Name: Floris van den Esschert

Internship or Thesis: Combined

Project Title: De-noising eye tracking data using machine learning.

A significant amount of research is being conducted in the area of eye-tracking. Over the last years, it has gained much traction in various fields, ranging from medical research, applications in gaming and HCI but also for business and enterprises to analyse and/or evaluate advertisements, UI/UX design, and shopping behaviour. One common problem among researchers using eye tracking is that variability in gaze position (noise) can pose problems when measuring relevant events that can be measured from such data (e.g. Fixation/saccades). Ideally, noise should be low enough such that it does not make events undetectable. Previous research has shown effective methods of event detection in situations with a high noise level, yet not much of this research has focussed on developing a tool which can automatically reduce random variability in eye-tracking signals without altering the signal significantly. In this paper I attempt to develop an automated tool which can remove or reduce noise such that it does not interfere with event detection, using various machine learning architectures. Given the low dimensional nature of the data, simple models should be able to realize this goal. The models that are going to be used are: PCA (Principal Component Analysis), LSTM (Long short-term memory), Hierarchical Hidden Markov Models (HHMM), simple feed forward neural network (FNN) and finally a Deep Convolutional Neural Network (DCNN). The models which are able to deal with this problem (if any) will be compared and a trade-off between computational time/complexity vs accuracy will determine the best model suited for this problem. In an ideal situation, the development of such a tool will provide a framework which can be built upon by

others or be widely implemented to help reduce labour of researchers and provide a clean dataset for data recorded through eye trackers.

Name Company or University: Utrecht University

UU Supervisors: Roy Hessels (AI), Ignace Hooge (ACP), (2nd supervisor) Chris P. Janssen (AI)